

SCOPE:

2. STANDARDS:

Sl. No.	IS	REMARKS
1	IS : 398 (Part I & II) of 1996	ACSR for overhead lines
2	IS 1778 of 1980	Reel & Drum for bare conductor
3	IS 4826 - 1968	Hot dip galvanized coating on Round steel wires

3. MAIN RESULTS

Element	% of composition
Manganese	0.40 - 1.10
Carbon	0.50 - 0.85
Phosphorus	0.05 max
Sulphur	0.05 max
Silicon	0.15 - 0.35

4. SIZE AND PROPERTIES:

5. TOLERANCES:

The following tolerances shall be permitted:

- iii. Tolerance on nominal diameter of galvanized steel wires: ± 2 (two) percent.

6. MODULUS OF ELASTICITY & CO-EFFICIENT OF LINEAR EXPANSION:

The values of the final modulus of elasticity and Co-efficient of linear expansion for ACSR shall be as given hereunder:

No. of Wires	Final Modulus of Elasticity GN/m ² (Practical)	Co-efficient of linear expansion/ ⁰ C.
ACSR30/7	80	$1/8 \times 10^{-6}$

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15.0 TESTREPORTS:

15.1 TYPE TEST:

TYPE TEST OF ACSR PANTHER CONDUCTOR

S. No	Particulars of Test	Clause No.
1.	Measurement of diameter of Individual Aluminium and Steel Wires	13.2 of IS- 398 (Part-II)
2.	Measurement of Lay Ratio/ Direction of Lay	13.8
3.	Breaking load of individual wire	13.3.1
4.	Ductility Test (Torsion/ Elongation)	13.4
5.	Wrapping Test (Aluminum and Steel)	13.5
6.	Resistance Test	13.6
7.	Galvanizing Test (Uniformity of Galvanizing and Mass of coating)	13.7
8.	Surface Condition Test	13.9
9.	Test for Ultimate Breaking Load on Stranded Conductor	13.10
10.	Stress Strain Test	13.11

15.2 The test reports of Routine/ Acceptance tests and checking of length etc, as specified in Clause no 13 shall be signed jointly by the supplier's representative and W&S&DCL'S representative.

15.1 PACKING & MARKING

- i) The conductor shall be wound on non-returnable drum strong enough and provided

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with lagging of adequate strength, constructed to protect the conductor against all displacement during transit, storage and subsequent handling and stringing operation in the field. The drum shall conform to IS: 1778-1980 as amended upto date and the dimensions shall be as per drum under column 9 of Table- 2 of the IS.

- ii) The drum shall be suitable for wheel mounting.
- iii) The general construction of drum shall be as shown in IS: 1778-1980. However, the drum shall be suitable for lifting off the conductor under controlled tension of the order of 300 kg minimum.
- iv) After application of bituminized and plastic paper protective lagging or circumferential battens of minimum 50mm thickness shall be provided suitably, in order to protect conductor from damage during transit in the event of breakage/detachment of the external protective lagging. The thickness of the external protective lagging or circumferential battens shall be sufficient to nail across grains as far as possible to the flange edges with at least one nail per end. The length of the nails shall be not less than twice the thickness of the battens. The nails shall not protrude above general surface and shall not expose sharp edges or allow the battens to be released due to corrosion.
- v) Outside the protective lagging, there shall be minimum two binders consisting of hoop iron or galvanized steel wire. Each protective lagging shall have recesses to accommodate hoop binders.
- vi) The conductor ends shall be properly sealed and secured with the hoop of "B" nails or bolts on the side of one of the flanges to avoid loosening of the conductor layers during transit and handling.

16.2 TOLERANCE QUANTITY:

A manufacturing tolerance upto (-)5% subject to maximum one standard drum length (age not each item of the order, for the last offered lot, will be allowed.

16.3 MARKING:

Each drum shall have the following information stenciled on it in indelible ink along with other essential details:

- a) Purchase Order number.
- b) Name and address of the consignee.
- c) Manufacturer's name or trademark.
- d) Drum number.
- e) Code name and size of the conductor.
- f) Length of the conductor.
- g) Gross weight of the drum.
- h) Weight of empty drum with protective lagging.
- i) Net weight of the conductor.
- j) Arrow marking for unwinding.
- k) Position of the conductor end.
- l) Lot number.

Before despatch, proper identification mark 'WBSEDCL' shall be engraved in each drum.

16.4 CONSTRUCTION OF DRUMS

- (a) All wooden components shall be manufactured out of seasoned soft wood free from defects that may materially weaken the component parts of the drums.

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(2) PARCLES

- (c) The flanges shall be of two ply construction with such ply at right angle of the other and nailed together. The nails shall be driven from the inside face of flanges, punched and then clinched on the outer face. There shall be at least 3 nail per plank of ply with maximum nail spacing 70- 75mm.
- (ii) There will be a slot in the flange to receive the inner end of the conductor; the entrance shall be in line with the periphery of the barrel.
- (c) Spindle hole saw shall be provided at the center of the middle planks of the pies and spindle planets with 100 mm diameter holes shall be fitted on either side of both the flanges.
- (d) DRUM AND SUPPORTS:
The end supports shall be securely fixed by nailing and may be disc or segmental type. The middle barrel support of the two ply construction of disc type will have 100 mm diameter concentric with the holes in flanges shall be provided at the centers of the barrel supports.
- (e) DRUM:
The wooden batons used for making the barrel of the conductor shall be segmental type. These shall be nailed to the barrel supports with at least two nails. The batons shall be closely butted and shall provide a round barrel with smooth surface. The edges of the batons shall be rounded or beveled to avoid damage to the conductor.
- (f) DRUM STUDS:
Barrel studs shall be used for the construction of drum. The flanges shall be holed and the barrel supports slotted to receive them. The barrel studs shall be threaded over a length on either end sufficient to accommodate washers, spacers, plates and nuts for fixing at the required spacing.
- (g) IRON COMPONENTS:
Nominally, the nuts on the studs shall stand proud of the flange. All the nails used on the inner surface of the flanges and the drum barrel shall be counter sunk at least 3 mm deep. The ends of barrel shall generally be finished with the top of the nuts.

16.5 PROTECTIVE ARRANGEMENT

25.5.5.A The inner side of the flanges and drum barrel surfaces shall be painted with bitumen based paint.

16.5.4.3 Before packing, cardboard of double corrugated or thick bituminised water proof bamboo paper shall be secured to the drum barrel and inside the flanges of drum by means of suitable adhesive materials. These protective wrappings and the adhesive materials used shall be of a quality which is not harmful to the conductor.

- i) After receiving the conductor, the exposed surface of the outer layer of the conductor shall be wrapped with water proof, thick, bituminised bamboo paper and also with thick plastic sheet to prevent the conductor from dirt, grit and damage during transport and handling.

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PRICE VARIATION:

The price shall be variable on the basis of prices of Aluminium and Steel as per CACMAI Price Variation Formula:

For every increase or decrease of Rs. 1,000/- (Rupees One thousand only) per M.T. in the prices of EC Grade Aluminium Rod and High Tensile Galvanised Steel Wire (HTGS) as per price circulated by CACMAI, the price of finished Conductors will be increased or decreased according to the CACMAI Price Variation Formula.

The price variation will be applicable as per the price prevailing on one month prior to the date of delivery of the particular lot.

BASE PRICE:

The prices (excluding Excise Duty, VAT, Central Sales Tax, if any) of EC Grade Aluminium Rod and High Tensile Galvanised Steel Wire (HTGS) as per price circulated by CACMAI as prevailing on the Base Date as per the Notice Inviting Tender will be the base prices of Aluminium and Steel.

Weights of Aluminium and Steel per Km. length of respective Conductors are given below:

Item No.	Description of materials	Weight in Kg. per Km.		
		Aluminium	Steel	Total
01	ACSR:200mm ² (PANTHER)	586.20	387.80	974.00

18.9 Documents to be submitted at the time of physical delivery at Consignee Stores:
The following documents are to be submitted by the vendors to the Consignee Stores at the time of dispatch to stores:-

- 18.1 Copy of Purchase order
- 18.2 Copy of Despatch Instruction
- 18.3 Inspection Test Certificate
- 18.4 Guaranteed Certificate
- 18.5 Proforma Invoice
- 18.6 Calculation Sheet for Price Variation on the basis of CACMAI as applicable with base date of Order
- 18.7 Seal list and packing list
- 18.8 Challan in triplicate
- 18.9 Way bill, if applicable

- (Signature of the Tenderer)

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1) SIZES & PROPERTIES OF ALUMINIUM CONDUCTOR (ALUMINISED) STEEL REINFORCED

NOMINAL ALUMINIUM AREA (mm ²)	STRAIGHTENING AND WELDING DIAMETER (mm) ALUMINIUM	STRAIGHTENING AND WELDING DIAMETER (mm) STEEL	SECTIONAL AREA OF ALUMINIUM (mm ²)	TOTAL SECTIONAL AREA (mm ²)	APPROXIMATE OVER-ALL DIAMETER (mm)	APPROXIMATE MASS (KG/KM)	CALCULATED RESISTANCE AT 20°C (N/CM ²)	APPROXIMATE CALCULATED BREAKING LOAD (KG)
400	24/1.00	7/3.25	244.1	261.5	21.06	374	100.3011	59.67

PROPERTIES OF ALUMINIUM WIRES USED IN THE CONSTRUCTION OF ALUMINIUM CONDUCTORS GALVANISED STEEL REINFORCED

NOMINAL DIAMETER (mm)	MIN	MAX	CROSS SECTIONAL AREA OF ALUMINIUM (mm ²)	MASS (KG/KM)	MIN STRETCH AT 20°C (N/CM ²)	MINIMUM LOAD (N/CM)	SPREADING LOAD AFTER STRAIGHTENING (KN)
3.20	2.97	3.07	7.869	20.11	4.075	17	4.11

PROPERTIES OF STEEL WIRES USED IN THE CONSTRUCTION OF ALUMINIUM CONDUCTOR STEEL REINFORCED

NOMINAL DIAMETER (mm)	MIN	MAX	CROSS SECTIONAL AREA OF STEEL (mm ²)	MASS (KG/KM)	MIN STRETCH AT 20°C (N/CM ²)	MINIMUM LOAD (N/CM)	SPREADING LOAD AFTER STRAIGHTENING (KN)
7.00	6.84	7.06	3.06	7.063	35.17	9.79	8.175

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